End-to-End Data Management Tool assembly for Norwegian Life Scientists

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Major funding agencies require the successful planning and implementation of Data Management. In addition to obtaining funding, good Data Management also makes research more efficient, facilitates collaboration and saves time and resources.
ELIXIR Norway and the Centre for Digital Life Norway have established an End-to-End tool assembly, covering the whole data life cycle from proposal to publication, providing practical, existing solutions, available to Norwegian Life Scientists to make their research **FAIR** (**F**indable, **A**ccessible, **I**nteroperable and **R**eusable).

Research question

Feide, LS LOGIN

Conceptualization
Grant Application
Data Management Plan
Budgeting

DSW

DATA STEWARDSHIP WIZARD

Questionnaire:
- Interactive Workflow
- Adapts to the stage of your project
- Provides information through references and tool overviews
- Tailored for Norwegian users with links to national resources

Data Management Plan:
- Automatically generated from filled questionnaire
- Can be exported to many formats including HTML, Word, LaTeX and machine readable JSON
- Export according Science Europe guidelines (e.g. used by RCN)

Developed by: Faculty of Information Technology CTU in Prague, DTL, GO FAIR, ÚOCHB IOCB Prague, ELIXIR Netherlands, ELIXIR Czech Republic

Experiments

nor seq
Direct upload from Core Facilities

Data Storage

Feide

Data Analysis

Computing
Statistics

Feide

Metadata tracking

Feide, LS LOGIN

Data integration
Sample & Data provenance

FAIRDOM SEEK

Supports ISA (Idea, Study, Assay) Format

Sample/Data provenance

Sharing of:
- Heterogeneous scientific research datasets
- Models or simulations
- Processes, SOPs and research outcomes
- Documents and presentations

Integration of JWS online for SBML models

Doi accessible snapshots for publication

National resource for Norwegian users

NeLS

Norwegian e-Infrastructure for Life Sciences

Storage Level I:
- Flexible storage with advanced access control
- Access via FEIDE via Browser or via SSH, SFTP/SCP
- Physical redundant hardware, 4 weeks protection against user error

Storage Level II:
- Structured, high volume, long term storage
- Secure national infrastructure - backuped and geo-replicated

Computing:
- Easy access to graphical bioinformatics tools from your Browser

TSD-quotas and bioinformatics pipelines for sensitive data

Support through our Helpdesk

Knowledge

Feide, LS LOGIN

Modelling

Preseve & Share

Data deposition

Feide, LS LOGIN

Direct depositition from NeLS
...coming soon

Established, community relevant data repositories
- Free of charge
- Long term availability
- Increase visibility of research

EGA for sensitive data
- Implemented data access committees
- Norwegian FEGA

PDBe, EMPIAR, Deposition Databases, Federated European Genome-phenome Archive Norway, BioImage Archive, IDR, ArrayExpress, ENA European Nucleotide Archive, BioModels, BioStudies, PRIDE, MetaboLights

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ELIXIR NORWAY

CENTRE FOR DIGITAL LIFE NORWAY

UNIVERSITY OF BERGEN
UiT The Arctic University of Norway
UiO : University of Oslo
NTNU Norwegian University of Science and Technology
NMBU Norwegian University of Life Sciences